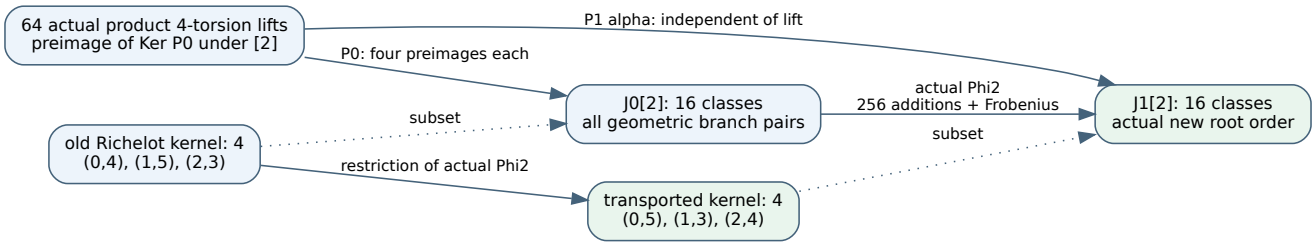
Geometric certificates for the seven-transfer chain

2026-11-11 · Chained ECDLP Transfers · CT21 progress checkpoint

Unsigned scientific standing. Root PCCba6bee binds producer-run native kernel/action certificates. PCC7409b3 binds six partial Lean helpers. This checkpoint accepts neither a complete next Psi corpus nor a solver or computational advantage. [pcc:unverified]



Actual kernel, beyond the abstract module candidates

The fixed F101 map Phi carries C0:$y^2=x^6+6x^4+13x^2+56$ to C1:$y^2=x^6+26x^4+42x^2+68$. Its degree49/ type(7,7) written proof is frozen in SEVEN_JACOBIAN_PROOF.md at root c1b9aa2. Existing native engines enumerate each selected elliptic7-kernel and apply the actual product pullback P0. The result has49 distinct divisor classes,all killed by7, all2401 additions and Frobenius closure. Observed eigenscalars5 and6 select one of the four earlier abstract candidates. This is the kernel, not all2401 points of J0[7]. [pcc:unverified]

Compute Phi on two-torsion without Jacobian halving

The old product gluing kernel has four points. Taking all four halves on each elliptic factor gives64 distinct product4-torsion lifts. Their P0 images cover all16 geometric J0[2] classes,four preimages each. P1 alpha images agree on every fiber and cover all16 target branch classes. All256 additions and Frobenius compatibility pass. No unknown DLP label occurs in the evaluator. [pcc:unverified]

The old cross-kernel maps to pairing(0,5),(1,3),(2,4) in the actual new root order. Determinant33 and rawtarget[78,66,56,53,91,29,88] select a concrete next Richelot candidate. Operational R1 and complete Psi acceptance are later stages;this diagram records the geometric action. [pcc:unverified]

Proof scope, reproduction and continuation

Reuse existing arithmetic and preserve field identity

The common flat $F101^6$ modulus is[33,46,2,18,56,12,1],gamma=T+s, $T^3+6T^2+13T+56=0$ and $s^2=2$. The adapter checks inverse/root equations, canonical tails and Frobenius$^3/^6$ against the existing tower. Actual new branch roots are derived from the elliptic7 image and remain in the OLD cubic coefficient field. No replacement field,Cantor or elliptic engine was written. [pcc:unverified]

Six conditional structural theorems

ChainTransport.lean proves kernel transport;the commuting square from factorization and surjective geometric doubling;composite norm28; rational half6159 on an order12317 group;full source reverse28;and target coordinate4. All six gate and local statement-binding checks pass with standard Lean foundations and zero trusted native uses. Total written source lemmas86. The first untyped-existential failure is retained. [pcc:unverified]

These helpers do not define finite fields,Picard groups,regular quotients,polarizations or unique elliptic subvarieties. The global geometric bridge is written;full formalization remains open. Geometric point groups have two-torsion;they are never assumed torsion-free. [pcc:unverified]

Concrete next acceptance

Transport the cross-kernel to a validated R1,then show the composite R1 Phi R0Hat kills all T0[2]. Quotient descent must give C=Psi[2], degree49 and both dual norms[7]. Require both complete12317-class rational domains,bijections,12317 paired-addition cases per direction, all109 original source scalars,reverse norm28/inverse74 and direct target coordinates([4]alpha1(P),0). Keep explicit source and elliptic controls,exceptional fibers and F101-Jacobian distinction. [pcc:unverified]

Exact sources and boundaries

Read SEVEN_TORSION_CHECKPOINT.md and CT21_PLAN.md. Evidence: evidence/seven-torsion-kernel-01.txt,seven-two-action-02.txt and frozen seven-torsion-source-snapshot-02.rs. Consumer starts2450815e2,branch research/chained-transfer-seven-torsion-20261111. Root Lean4.35.0: tools/lean/proof-gate.sh with ChainTransport.lean;native Rust1.99.1. The published Phi rung is ready cryptanalysisPR1484. [pcc:unverified]

No canonical atlas graph is promoted:KB/ecdb unavailable and atlas8edec422a vault locked. S3 admission/livebudget block benchmarks,solvers and paid work. Root ledger fork/archive-clock gates persist. Same-prime coefficient descent supplies no cross-prime map,novelty or measured speed claim. [pcc:unverified]